Data Description: Forecasting S&P 500 Realized Volatility (v5.2)

Every file, every column, and the role it plays in the project

Companion to Project Proposal v5.2 · Advanced Machine Learning (QFGB-890H-001) · Fordham MSQF · Fall 2026

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How to read this: One section per file, headed by the exact file name, with a short intro (source, dates, frequency, units, caveats) and a table that lists every column with a one-line description and its role. Roles follow the v5.2 proposal: main target, secondary target, feature set A, B, C or D, cross-check/robustness, variance-swap strike, QC/audit only, or **not used in v5.2**. Where v5.2 does not name a column, the role is marked "not used in v5.2" or "optional". Descriptions come from the files themselves and their READMEs and data dictionaries.

Role labels at a glance

Label	Meaning in v5.2
Main target	Risk Lab 5-minute SPY realized variance, $\mathrm{RV}_t = (\texttt{rv5_trade_vol})^2/252$, averaged over $h = 1, 5, 22$ days
Secondary target	VOLARE ES rv5 (already a daily variance), 2010–2026; used for RQ3 and the variance-swap check
Set A	HAR lags: daily, weekly (5-day), monthly (22-day) RV averages; optional 15-min RV and QMLE. Used by HAR, elastic net and random forest
Set B	A + market: negative-return averages, Parkinson range, log VIX, IV surface (ATM/skew/slope), VRP, rates; short-dated IV/VVIX when available
Set C	B + macro/stress: ADS, EPU, news sentiment, GPR, CISS, NFCI, STLFSI4, Hang Seng, calendar flags
Set D	C + ES measures: continuous/jump averages, ± semivariances, √quarticity×RV, realized kernel, overnight variance max(ES−SPY, 0); from 2010
Variance-swap strike	$K_t \approx (\mathrm{VIX}_t/100)^2$ (i.e. $\mathrm{VIX}^2/252$ per day), the market benchmark in the economic check, not a forecasting model
Cross-check / robustness	Used only in the robustness checks listed in v5.2 Section 6
QC / audit only	Helper or audit column; not a model input
Not used in v5.2	Present in the file but not part of the v5.2 design

1. risklab_SPY_daily_annualized_vol_1996-2026.csv

Source: Dacheng Xiu's Risk Lab (Chicago Booth), SPY (PERMNO 84398), research use only. **Dates:** 2 January 1996 – 5 October 2026, 7,707 daily rows, one per NYSE session (33 sessions since 2005 are missing, including 5–9 February 2018). **Frequency:** daily, regular trading session only (no overnight). **Units:** every column ending in _vol is an **annualized volatility in decimals** (0.117 = 11.7%). Daily variance = vol² ÷ 252. No NaNs in the file.

Which column is the target? v5.2 uses **rv5_trade_vol** (trade-based). The quote-based series have outliers (up to 893% annualized, max 8.93 in the file) while the trade series ranges from 2.0% to 121.6%. The 15-minute RV and QMLE (trade) are the "optional 15-min RV and QMLE" in set A, and also robustness targets.

Column	What it is (one line)	Units	Role in v5.2
date	NYSE session date (YYYY-MM-DD)	date	Index / alignment key
qmle_trade_vol	Noise-robust QMLE volatility estimate from SPY trades	ann. vol, decimal	Set A (optional); robustness ("Risk Lab 15-min and QMLE")
qmle_trade_q	Lag order q of the MA noise model chosen by the QMLE (0–12)	integer	QC / audit only
qmle_trade_se	Standard error of qmle_trade_vol	decimal	QC / audit only
rv5_trade_vol	5-minute subsampled RV from trades	ann. vol, decimal	**Main target**; source of the set-A HAR lags

Column	What it is (one line)	Units	Role in v5.2
rv15_trade_vol	15-minute subsampled RV from trades	ann. vol, decimal	Set A (optional); robustness
qmle_quote_vol	QMLE volatility from SPY quotes	ann. vol, decimal	Not used in v5.2 (quote outliers)
qmle_quote_q	MA lag order for the quote QMLE	integer	Not used in v5.2
qmle_quote_se	Standard error of qmle_quote_vol	decimal	Not used in v5.2
rv5_quote_vol	5-minute RV from quotes	ann. vol, decimal	Not used in v5.2 (outliers)
rv15_quote_vol	15-minute RV from quotes	ann. vol, decimal	Not used in v5.2

> **Heads-up:** Also used outside this file: rv5_trade_vol is the SPY side of the set-D **overnight variance** max(ES rv5 − SPY RV, 0), and the target for which VOLARE ES is *never* used as a stand-in on missing days.

2. VOLARE_ES_realized_measures_2009-2026.csv

Source: VOLARE (Cipollini et al., 2026), volare.unime.it, downloaded 4 October 2026, research use; E-mini S&P 500 futures (ES), built from 1-second mid-quotes. **Dates:** 30 September 2009 – 30 September 2026, 4,397 rows × 32 columns. **Frequency:** daily; each date covers the full Globex session (18:00 ET the previous evening to 17:00 ET), so it **includes the overnight session** (ES rv5 ≈ 95% of close-to-close variance vs ≈ 56% for SPY RV). **Units:** all measures are **daily variances of log returns in decimals** (not annualized), except prices. **Caveats:** no NaN, zero or negative values; no NYSE session missing; 121 extra holiday rows, which are added to the next NYSE day (dropping them is a robustness check). Roll dates are not flagged. Refreshed monthly. In v5.2 ES is used **from 2010** (development 2010–2017).

Suffixes: **1** = 1-minute returns, **5** = 5-minute, **_ss** = 5-minute subsampled. There is no jump column: v5.2 builds $J_t = \max(\mathrm{rv5} - \mathrm{bv5}, 0)$ and the continuous part $C_t = \mathrm{rv5} - J_t$.

Column	What it is (one line)	Units	Role in v5.2
date	Trading date (CME trade date)	date	Index / alignment key
symbol	Instrument code; always ES	text	QC / audit only
asset_type	Asset class; always futures	text	QC / audit only
open_price, close_price, high_price, low_price	Front-contract open, close, high and low	index points	Not used in v5.2 (used in QC to infer the session)
pr	Parkinson high–low range variance	daily var.	Not used in v5.2
gkr	Garman–Klass range variance	daily var.	Not used in v5.2
rr5	5-minute realized range	daily var.	Not used in v5.2
rv1	Realized variance, 1-minute returns	daily var.	Robustness (“ES rv1”)
rv5	Realized variance, 5-minute returns	daily var.	**Secondary target**; Set D (continuous/jump split, overnight variance)
rv5_ss	5-minute subsampled RV	daily var.	Not used in v5.2
bv1	Bipower variation (jump-robust), 1-minute	daily var.	Not used in v5.2
bv5	Bipower variation, 5-minute	daily var.	Set D (jump J = max(rv5 − bv5, 0); continuous part)
bv5_ss	Bipower variation, 5-minute subsampled	daily var.	Not used in v5.2
rsp1, rsn1	Positive / negative semivariance, 1-minute (rsp + rsn = rv)	daily var.	Not used in v5.2
rsp5, rsn5	Positive / negative semivariance, 5-minute	daily var.	Set D (± semivariances)
rsp5_ss, rsn5_ss	Subsampled 5-minute semivariances	daily var.	Not used in v5.2
medrv1, medrv5, medrv5_ss	MedRV jump-robust estimator (1-min, 5-min, subsampled)	daily var.	Not used in v5.2

Column	What it is (one line)	Units	Role in v5.2
minrv1, minrv5, minrv5_ss	MinRV jump-robust estimator	daily var.	Not used in v5.2
rk	Realized kernel on 1-second returns (noise-robust)	daily var.	Set D (realized kernel); robustness ("kernel")
rq1	Realized quarticity, 1-minute, $(m/3)\Sigma r^4$	variance²	Not used in v5.2
rq5	Realized quarticity, 5-minute	variance²	Set D (√quarticity × RV)
rq5_ss	Realized quarticity, 5-minute subsampled	variance²	Not used in v5.2

3. oxfordman_SPX_2000-01-03_2022-02-25.csv

Source: Oxford-Man Institute Realized Library, archived mirror (the library is discontinued); S&P 500 index (.SPX) rows only. **Dates:** 3 January 2000 – 25 February 2022, 5,552 rows × 19 columns, no NaNs (22 NYSE sessions missing, mostly 2018–2022). **Frequency:** daily, cash-index trading hours. **Units:** realized measures are **daily variances in decimals**; prices in index points. **Caveats:** open_time/close_time appear in two clocks (93000/160000 and 83000/150000), so do not infer hours from them; no quarticity. **Role:** cross-check of Risk Lab rv5 (correlation 0.968 over 2005–2022) and the source of **pre-2010 jump features** in the robustness check.

Column	What it is (one line)	Units	Role in v5.2
date	Trading date	date	Index / alignment key
rv5	Realized variance, 5-minute returns	daily var.	Cross-check vs Risk Lab rv5
rv5_ss	5-minute subsampled RV	daily var.	Not used in v5.2
rv10, rv10_ss	10-minute RV and its subsampled version	daily var.	Not used in v5.2
bv	Bipower variation (jump-robust)	daily var.	Robustness (pre-2010 jump features)
bv_ss	Subsampled bipower variation	daily var.	Not used in v5.2
rsv	Negative realized semivariance	daily var.	Robustness (pre-2010 semivariance)
rsv_ss	Subsampled negative semivariance	daily var.	Not used in v5.2
medrv	MedRV jump-robust estimator	daily var.	Not used in v5.2
rk_parzen	Realized kernel, Parzen weights	daily var.	Not used in v5.2
rk_twoscale	Two-scale realized variance	daily var.	Not used in v5.2
rk_th2	Realized kernel, Tukey–Hanning(2) weights	daily var.	Not used in v5.2
open_price, close_price	Index open and close	index points	Not used in v5.2
open_to_close	Open-to-close log return	decimal	Not used in v5.2
open_time, close_time	Time of first / last observation (HHMMSS, mixed clocks)	time	QC / audit only (unreliable)
nobs	Number of intraday observations used	count	QC / audit only

4. features_daily.csv

Source: the team's free supplementary file, built 7 October 2026 from public endpoints (Yahoo Finance, FRED, Philadelphia Fed, Ken French, ECB, SF Fed, Iacoviello, Cboe), with DATA_DICTIONARY.md and QC_REPORT.md. **Dates:** every NYSE session 2 January 1990 – 6 October 2026, 9,258 rows, 51 data columns plus date. **Frequency:** daily (weekly series carried forward between releases). **Alignment:** daily series sit on their *observation* date and are **not forward-filled** (NaN = no value that day or before the series starts). For forecasting, v5.2 shifts each column by its publication lag (box below; lags also noted in the Role column).

Publication lags applied in v5.2: market closes, Treasury rates, Hang Seng (closes before the NYSE open) and Cboe indices: used as of day t · EPU **+1 trading day** · CISS **+2** · SF Fed news sentiment and GPR **+5** (weekly) · NFCI/STLFSI4 already placed on their **release date** (week-ending Friday + 6 days) · ADS: **real-time vintage** from December 2008 (the current vintage is heavily revised) · Fama-French factors: about **1 month** behind, so not usable in real time.

4.1 Calendar and S&P 500 (Yahoo ^GSPC)

Column	What it is (one line)	Units	Role in v5.2
date	NYSE session date	date	Index / alignment key
nyse_early_close_flag	1 = scheduled early close (usually 13:00 ET)	0/1	Set C (calendar flags)
nyse_close_time_et	Scheduled close time	HH:MM ET	QC / audit only
spx_open_pts	S&P 500 open (stale = previous close for ~1990–2007, parts of 2011–13)	points	Not used in v5.2 directly
spx_high_pts, spx_low_pts	Daily high and low	points	Set B (inputs to Parkinson range)
spx_close_pts	Daily close (price index)	points	Set B (input to returns)
spx_volume_shares	Composite volume	shares	Not used in v5.2
spx_logret_cc_pct	Close-to-close log return, $100 \cdot \ln(C_t/C_{t-1})$	percent	Set B (negative-return averages)
spx_parkinson_var_daily	Parkinson range variance $(\ln H/L)^2/(4 \ln 2)$	daily var.	Set B (Parkinson range)
spx_garman_klass_var_daily	Garman–Klass variance; NaN when the open is stale (3,699 NaN)	daily var.	Not used in v5.2
spx_open_equals_prev_close_flag	1 = Yahoo open equals previous close (stale)	0/1	QC / audit only

4.2 Hang Seng (Yahoo ^HSI)

HK closes at 04:00 ET, so day-t values are known before the NYSE open. NaN on NYSE days when HKEX was closed (411 rows).

Column	What it is (one line)	Units	Role in v5.2
hsi_open_pts, hsi_high_pts, hsi_low_pts, hsi_close_pts	Hang Seng daily OHLC	points (HKD)	Not used directly (inputs)
hsi_logret_cc_pct	Close-to-close log return over HK sessions	percent	Input to the squared return
hsi_sq_logret_pct2	Squared Hang Seng return (overnight Asian shock)	percent^2	Set C (Hang Seng)
hsi_traded_same_date_flag	1 = HKEX traded on that date	0/1	QC / audit only

4.3 Rates and credit (FRED)

Column	What it is (one line)	Units	Role in v5.2
t10y3m_spread_pctpts	10-year minus 3-month Treasury yield (term slope); NaN on bond holidays	pct. points	Set B (rates)
dgs10_yield_pct	10-year Treasury yield	percent	Set B (rates)
hy_oas_pctpts	High-yield option-adjusted spread; only from 9 Oct 2023 (FRED licensing)	pct. points	Not used in v5.2 (too short)

4.4 Uncertainty, financial conditions and business conditions

Column	What it is (one line)	Units	Role in v5.2
epu_daily_index	Baker–Bloom–Davis daily news-based Economic Policy Uncertainty	index	Set C (EPU), lag +1 day
nfci_index_asof_release	Chicago Fed NFCI, placed on release date (+ = tighter); weekly, re-estimated each week	std. dev.	Set C (NFCI)

Column	What it is (one line)	Units	Role in v5.2
nfci_obs_week_ending	Which NFCI week the row holds	date	QC / audit only
stlfsi4_index_asof_release	St. Louis Fed Financial Stress Index; weekly, release-aligned; NaN before 6 Jan 1994	index (0 = normal)	Set C (STLFSI4)
stlfsi4_obs_week_ending	Which STLFSI4 week the row holds	date	QC / audit only
ads_index_current_vintage	ADS business-conditions index, latest vintage (heavily revised)	std. dev.	Not used for test forecasts (look-ahead risk); per v4, pre-Dec-2008 training rows only, lagged
ads_index_realtime_latest	ADS as known in real time on day t; from 8 Dec 2008	std. dev.	Set C (ADS real-time vintage)
ads_index_realtime_obs_date	Observation date the real-time ADS value refers to	date	QC / audit only
ciss_us_new_index	ECB New CISS for the United States (systemic stress)	index 0–1	Set C (CISS), lag +2
ciss_ea_new_index	ECB New CISS for the euro area	index 0–1	Not used in v5.2 (alternative)
sf_news_sentiment_index	SF Fed daily news sentiment (+ = positive); weekly updates	index	Set C (news sentiment), lag +5
ai_gpr_index	Iacoviello–Tong AI geopolitical risk index; monthly updates, revised	index (mean 100)	Set C (GPR); lag ~1 month or use classic GPR
ai_gpr_threats_index, ai_gpr_acts_index	AI-GPR sub-indices: threats and acts	index	Not used in v5.2
gpr_keyword_3paper_index	Keyword GPR on the same three newspapers	index	Not used in v5.2
gprd_classic_index	Caldara–Iacoviello classic daily GPR; weekly updates	index	Set C (GPR alternative), lag +5

4.5 Fama-French factors (Ken French)

Ken French Data Library, FF5 (2×3) and momentum, in **percent per day**, 1990 – 31 August 2026 (later rows NaN). Updated monthly with about a 1-month lag.

Column	What it is (one line)	Units	Role in v5.2
ff_mkt_rf_pct	Market excess return (Mkt − RF)	% per day	Not used in v5.2 (~1-month lag)
ff_smb_pct	Size factor, small minus big	% per day	Not used in v5.2
ff_hml_pct	Value factor, high minus low book-to-market	% per day	Not used in v5.2
ff_rmw_pct	Profitability factor, robust minus weak	% per day	Not used in v5.2
ff_cma_pct	Investment factor, conservative minus aggressive	% per day	Not used in v5.2
ff_rf_pct	Risk-free rate (1-month T-bill)	% per day	Not used in v5.2
ff_mom_pct	Momentum, past winners minus losers	% per day	Not used in v5.2

4.6 Other Cboe volatility indices

Column	What it is (one line)	Units	Role in v5.2
vvix_close	VVIX, implied vol of the VIX (vol of vol); from Mar 2006, gaps in 2006	index points	Set B (VVIX when available; from 2007)
vxn_close	VXN, Nasdaq-100 implied vol; from Feb 2001 (FRED backfill pre-2009)	index points	Set B (sector/cross-market IV, optional)
vxd_close	VXD, DJIA implied vol; from Oct 1997	index points	Set B (cross-market IV, optional)
rvx_close	RVX, Russell 2000 implied vol; from Jan 2004	index points	Set B (cross-market IV, optional)

Column	What it is (one line)	Units	Role in v5.2
vxn_source, vxd_source, rvx_source	Which source filled the value (Cboe CDN or FRED)	text	QC / audit only

5. vol_indices.mat

Source: Cboe daily index history files (free), cleaned by OptionPricingData.m (Option Pricing Data/data/clean). **Contents:** one MATLAB timetable **V** (row times d), 9,289 rows, 2 January 1990 – 1 October 2026 (union of the six indices' dates). **Units:** VIX family in **percentage points** (18.1 = 18.1%); SKEW as an index level. **Caveats:** NaN before each index existed, and VIX and SKEW are NaN on the last row (1 Oct 2026; their files end 30 Sep).

Column	What it is (one line)	Units	Role in v5.2
VIX	30-day S&P 500 implied vol; from 1990	vol points	Set B (log VIX, VRP); **variance-swap strike** $K_t \approx (\mathrm{VIX}_t/100)^2$
VIX9D	9-day implied vol; from 4 Jan 2011	vol points	Set B (short-dated IV, from Jan 2011)
VIX3M	3-month implied vol; from 18 Sep 2009	vol points	Set B (term structure, when available)
VIX6M	6-month implied vol; from 2 Jan 2008	vol points	Set B (term structure)
VIX1Y	1-year implied vol; from 3 Jan 2007	vol points	Set B (term structure)
SKEW	Cboe SKEW, price of tail (crash) risk; 100 = no premium; 101–183 in data	index level	Set B (implied-vol features)

Heads-up: $\mathrm{VIX}^2/252$ is **not** a forecasting model in v5.2 (Section 4.6). VIX² enters only as the market strike in the variance-swap check, $\Pi_t = K_t - 252 \cdot \mathrm{RV}_{t+1:t+22}$.

6. iv_surface.mat

Source: Bloomberg terminal, SPX implied-volatility surface (fields <tenor>_IMPVOL_<moneyness>%MNY_DF), cleaned by OptionPricingData.m Section 5. **Dates:** 3 January 2005 – 1 October 2026, 5,471 trading days. **Units:** implied vol in **decimals** (0.181 = 18.1%); in the file values range 0.052–0.857. **Caveats:** 658 of 229,782 values (0.29%) are NaN, all from the 24-month tenor before 18 May 2005. The surface start (January 2005) is why the v5.2 main sample starts in 2005.

Column	What it is (one line)	Units	Role in v5.2
IV	Implied vol array, IV(t, i, j) = day t, moneyness i, tenor j; 5,471 × 7 × 6	decimal	Set B: ATM level, skew and term slope
ivDates	Trading dates for the rows of IV; 5,471 × 1	datetime	Index / alignment key
moneyness	Strike ÷ spot: 0.90, 0.95, 0.975, 1.00 (ATM, index 4), 1.025, 1.05, 1.10	ratio	Defines ATM (1.00) and skew points
tenorYrs	Years to expiry: 1/12, 2/12, 3/12, 6/12, 1, 2 (1M–24M)	years	Defines the term slope

Per v4, the set-B features built from it are ATM 1M, 3M and 1Y implied vol, **skew** = IV(90%) − IV(110%) at 1 month, and **term slope** = IV(1Y) − IV(1M) at the money. The 2Y tenor has the most gaps and is used only for the term-slope check.

Summary: which columns feed each feature set

File keys: risklab_SPY…csv = risklab_SPY_daily_annualized_vol_1996-2026.csv · VOLARE_ES…csv = VOLARE_ES_realized_measures_2009-2026.csv · oxfordman_SPX…csv = oxfordman_SPX_2000-01-03_2022-02-25.csv · features_daily.csv · vol_indices.mat · iv_surface.mat.

Set / role	Columns (file)	Start
Main target	rv5_trade_vol (risklab_SPY…csv) → $\mathrm{RV}_t = \mathrm{vol}^2/252$, averaged over 1, 5, 22 days	Jan 2005 (Risk Lab from 1996)
Secondary target	rv5 (VOLARE_ES…csv), holiday rows added to the next NYSE day	Jan 2010 (file from 30 Sep 2009)
A: HAR lags (HAR, EN, RF)	Daily, weekly and monthly averages of rv5_trade_vol²/252 (risklab_SPY…csv); optional rv15_trade_vol and qmle_trade_vol (same file). For ES-target runs the same lags of ES rv5	Jan 2005
B: + market (EN, RF)	spx_logret_cc_pct → negative-return averages; spx_parkinson_var_daily (from spx_high/low_pts); t10y3m_spread_pctpts, dgs10_yield_pct; vvix_close, vxn_close, vxd_close, rvx_close (features_daily.csv) · VIX → log VIX and VRP; VIX9D, VIX3M, VIX6M, VIX1Y, SKEW (vol_indices.mat) · IV → ATM, skew, term slope (iv_surface.mat)	Jan 2005; VVIX from 2007; VIX9D from Jan 2011
C: + macro/stress (EN, RF)	ads_index_realtime_latest, epu_daily_index, sf_news_sentiment_index, ai_gpr_index / gprd_classic_index, ciss_us_new_index, nfci_index_asof_release, stlfsi4_index_asof_release, hsi_sq_logret_pct2, nyse_early_close_flag (features_daily.csv)	Jan 2005; real-time ADS from Dec 2008
D: + ES measures (EN, RF)	rv5, bv5 → continuous/jump averages; rsp5, rsn5 → ± semivariances; rq5 → √quarticity × RV; rk → realized kernel (VOLARE_ES…csv); overnight variance max(ES rv5 − SPY RV, 0) (VOLARE_ES…csv + risklab_SPY…csv)	Jan 2010
Variance-swap strike	VIX (vol_indices.mat) → $K_t \approx (\mathrm{VIX}_t/100)^2$; compared with 252 × the next-22-day RV	Jan 2005 (SPY), Jan 2010 (ES)
Cross-check / robustness	rv5, bv, rsv (oxfordman_SPX…csv); rv15_trade_vol, qmle_trade_vol (risklab_SPY…csv); rv1, rk (VOLARE_ES…csv); pre-2005 Risk Lab for the VIX-only 1996 start	2000–2022 (Oxford-Man)
Not used in v5.2	Quote-based Risk Lab series; most VOLARE alternative estimators (_ss, 1-min bv/rs/rq, medrv, minrv, range); Garman–Klass, volume, HY OAS, EA CISS, GPR sub-indices; Fama-French factors (features_daily.csv)	—